

SARABDEEP SINGH BILKHU

Sarabdeep.singh.bilkhu@gmail.com | +91 70424-32513 | Jalandhar, Punjab, India
LinkedIn: [linkedin.com/in/sarabdeep-singh-bilkhu](https://www.linkedin.com/in/sarabdeep-singh-bilkhu) | Portfolio: sarabdeepsbilkhu.github.io

Profile

Computer Science Engineering student interested in software systems, backend development, developer tooling, and intelligent computing. Experienced in building full-stack applications, automation frameworks, and real-time systems using Python and modern web technologies. Seeking to advance technical expertise through research and higher studies.

Educational Qualifications

April, 2024 <i>Rajgomaal, Jalandhar</i>	Higher Secondary Certificate S.T.S. World School
Expected June 2028 <i>Phagwara, Punjab</i>	Bachelor of Engineering – Computer Science Lovely Professional University, Punjab
CGPA: 9.07 / 10.00 (through Semester III)	

Research Publications

R. Gupta, S. S. Bilkhu, R. Khurana and A. Kumar,
“**Enhancing Automated Interactions via Hybrid Gesture Recognition,**”
2025 International Conference on Intelligent and Secure Engineering Solutions (CISES),
Greater Noida, India, pp. 1060–1065, 2025.
DOI: [10.1109/CISES66934.2025.11265624](https://doi.org/10.1109/CISES66934.2025.11265624)

Projects

AlgoTrace

Jan, 2026

- Built a proxy-based execution engine for execution replay and visualization.
- Implemented interactive visualizations for arrays, trees, and graph algorithms.
- Integrated Monaco Editor to support code execution and algorithm exploration.

Backend-in-a-Box

Feb, 2026

- Built a schema-driven FastAPI backend generator using YAML.
- Automated generation of models, REST APIs, validation schemas, and migrations from YAML specifications.
- Reduced development effort through reusable code generation workflows.

Technical Skills

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|---|---|
| • Languages: C, C++, Python, SQL | • Concepts: Data Structures, OOP |
| • Web: React, Node.js | • Tools: Git, AWS |
| • Databases: MySQL, Firebase, Supabase | • AI: Machine Learning, Computer Vision, NumPy, Pandas |

Additional Information

- Languages: English (Fluent), Hindi (Fluent), Punjabi (Fluent), Japanese (Beginner).
- Achieved 4 kyu rank on Codewars.
- Received Top 10 recognition at a university project exhibition.
- NCC alumnus with leadership, discipline, and teamwork experience.
- Presented a Gesture Recognition project at CU CAB Meet 2025.